Units of area

1 Consider the following units of measure for area:

- Hectares
- Square centimetres
- Square metres
- Square kilometres

Choose the most appropriate unit of measure for the following areas:

a Postage stamp

b Bathroom wall

c Australia

d Matchbox

e Exercise book

f Street

g School playground

h Pencil case

i Japan

j Cabinet

k City

l Television

m Kitchen

n Farm

o Classroom

p Netball court

q Table

r National park

s Door

2 Convert the following to the unit indicated:

a 3 m^2 to square centimetres

b 7 m^2 to square centimetres

c 0.56 km^2 to square metres

d 810 mm^2 to square centimetres

e $34\,000 \text{ mm}^2$ to square metres

f $790\,000\,000 \text{ cm}^2$ to square kilometres

g 6 ha to square metres

h 10.4 ha to square metres

i $20\,000 \text{ m}^2$ to hectares

j $54\,800 \text{ m}^2$ to hectares

3 Find the area of a square of side 290 cm, in square metres.

Applications

4 A park has an area measuring 19 acres. Given that 1 acre is approximately 4000 m^2, find the area of the park in square metres.

5 Rectangular farms around Australia were [redacted]red and their dimensions, in metres, are recorded in the following table:

a Find the area of each farm in square metres.

b Which farms have an area of more than $1 \mathrm{~ha}$?

c Which farms have an area of less than $1 \mathrm{~ha}$?

d Which farm has an area of exactly $1 \mathrm{~ha}$?

Farm	Length, m	Width, m	Area, m^2
1	200	100	
2	350	21	
3	100	24	
4	400	40	
5	100	100	

6 A property covers an area of $4.5 \mathrm{~ha}$. Given that $1 \mathrm{~ha}$ is $10\,000 \mathrm{~m}^2$, calculate the following:

a The area of the property in square metres.

b The area of the property in acres, given that 1 acre is approximately $4000 \mathrm{~m}^2$.

7 A large cattle station covers an area of 320 acres. Given that 1 acre is approximately $0.4 \mathrm{~ha}$, and $1 \mathrm{~ha}$ is $10\,000 \mathrm{~m}^2$, calculate the following:

a The area of the cattle station in hectares.

b The area of the cattle station in square kilometres.

8 A farm covers an area of $51\,000 \mathrm{~m}^2$. Given that $1 \mathrm{~ha}$ is $10\,000 \mathrm{~m}^2$, and 1 acre is approximately $4000 \mathrm{~m}^2$, calculate the following:

a The area of the farm in hectares.

b The area of the farm in acres. Round your answer to two decimal places.

9 Bob owns a paddock with an area of 290 acres. He wants to advertise his eggs as free range, with no more than 400 chickens per hectare. Given that 1 acre is approximately $0.4 \mathrm{~ha}$, find the largest amount of chickens that Bob can keep in his paddock.

10 A farm covers an area measuring 340 acres. Given that 1 acre is approximately $0.4 \mathrm{~ha}$, and $1 \mathrm{~ha}$ is $10\,000 \mathrm{~m}^2$, calculate the following:

a The area of the farm in hectares.

b The area of the farm in square metres.

11 American football is played on a field with an area of $12\,000 \mathrm{~yd}^2$.

a Given that $1 \mathrm{~yd}^2$ is approximately $0.836 \mathrm{~m}^2$, find the area of an American football field in square metres.

b The MCG oval has an area of $17\,700\ \text{m}^2$ [redacted] much smaller than the MCG oval is an American football field? Give your answer in square metres.

12 Manhattan, the most densely populated borough of New York City, has an area of $22.8\ \text{mi}^2$. If $1\ \text{mi}^2$ is approximately $2.6\ \text{km}^2$, how large is Manhattan in square kilometres?

13 The state of California is roughly $164\,000\ \text{mi}^2$ in area. Given that $1\ \text{mi}^2$ is approximately $2.6\ \text{km}^2$:

a Find the area of California in square kilometres.

b The state of New South Wales has an area of approximately $809\,000\ \text{km}^2$. How much larger is New South Wales than California?

14 An official NBA court covers $4700\ \text{ft}^2$ in area. Given that $1\ \text{m}^2$ is approximately $10.8\ \text{ft}^2$, find the area of an official basketball court in square metres. Round your answer to three decimal places.